Software Engineering Research Group MSc Thesis Style

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Software Engineering Research Group MSc Thesis Style

THESIS

submitted in partial fulfillment of the
requirements for the degree of

MASTER OF SCIENCE

in

COMPUTER SCIENCE

by

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born in Weert, the Netherlands

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Cover picture: A “random” maze generated in postscript.

Software Engineering Research Group MSc Thesis Style

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Abstract

This document describes the standard thesis style for the Software Engineering department at Delft University of Technology. The document and it's source are an example of the use of the standard LaTeX style file. In addition the final appendix to this document contains a number of requirements and guidelines for writing a Software Engineering MSc thesis.

Your thesis should either employ this style or follow it closely.

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Preface

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Leon Moonen
Delft, the Netherlands
April 8, 2026

Contents

Preface **iii**

Contents **v**

List of Figures **vii**

1 Introduction **1**
1.1 Terminology 1
1.2 Research Question(s) 1

2 Background & Theory **5**
2.1 Operator Learning 5
2.2 Fourier Neural Operator 5
2.3 Convolutional Neural Operator 6
2.4 Mesh-free Operator Learning and geo-FNO 6

3 Experimental Setup **7**
3.1 Datasets 7
3.2 Evaluation Metrics 9
3.3 Training Details 10
3.4 Resolution Transfer Protocol 10

4 Results **11**
4.1 Part I: Discretization Invariance Study 11
4.2 Part II: Non-uniform Meshes and geo-FNO 12
4.3 Writing note 13

5 Related Work **15**
5.1 Related part I 15
5.2 Related part II 15
5.3 Related part III 16

6 Conclusions and Future Work **17**
6.1 Contributions 17
6.2 Conclusions 18
6.3 Discussion/Reflection 18
6.4 Future work 19

Bibliography **21**

A Glossary **23**

B Requirements and Guidelines **25**
B.1 Requirements 25
B.2 Guidelines 26

List of Figures

3.1 Example of an input diffusion coefficient field $a(x,y)$ and the resulting pressure field $u(x,y)$ with Dirichlet boundary conditions. 8

Chapter 1

Introduction

Here I am just adding something new so that I can tell whether the compile is new.

This chapter introduces all aspects that form the context and motivations of the graduate project. We will describe ...

1.1 Terminology

The term *reverse engineering* is defined by Chikofsky and Cross as "*The process of analyzing a subject system with two goals in mind: (1) to identify the system's components and their interrelationships; and, (2) to create representations of the system in another form or at a higher level of abstraction*" [**?**].

Note that this Terminology section is not a required section but merely an example.

1.2 Research Question(s)

The research question investigated in this thesis is ...

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Chapter 2

Background & Theory

This chapter introduces the theoretical background needed for the remainder of the thesis. It first places operator learning in context with more classical learning setups and then discusses the neural operator architectures that are central to this work.

2.1 Operator Learning

This section introduces operator learning as a mapping between functions, rather than the more familiar vector-to-vector setting. It should explain why this perspective is useful for PDE-governed physical systems and introduce the CDE concept that motivates resolution-aware learning.

2.1.1 Vector-to-vector versus function-to-function mapping

Contrast conventional supervised learning with operator learning and explain why PDE solution operators are naturally formulated in the function-to-function setting.

2.1.2 Continuous-domain perspective

Introduce the continuous-domain view and connect it to the notion of discretization invariance used throughout the thesis.

2.2 Fourier Neural Operator

This section should explain the FNO architecture and the spectral convolution mechanism on which it is based.

2.2.1 Architecture

Describe the lifting layer, Fourier layers, spectral convolution, non-linear transformations, and projection back to the target field.

2.2.2 Strengths

Discuss the global receptive field, computational efficiency through FFT-based convolutions, and the suitability of FNO for smooth solution operators.

2.2.3 Weaknesses

Discuss spectral truncation, sensitivity to resolution changes, and the reliance on regular grids.

2.3 Convolutional Neural Operator

This section should explain the CNO architecture and the inductive biases that differentiate it from FNO.

2.3.1 Architecture

Describe the encoder-decoder style structure, convolutional blocks, and the way multi-scale processing is used.

2.3.2 Bandlimited assumption

Explain the bandlimited assumption and why it can stabilize learning and support more consistent behavior across discretizations.

2.3.3 Local versus global inductive bias

Compare the locality of convolutional operators with the global spectral bias of FNO and connect this to the later experiments.

2.4 Mesh-free Operator Learning and geo-FNO

This section motivates operator learning on irregular domains and introduces geo-FNO as the architecture used for this purpose.

2.4.1 Motivation for irregular domains

Explain why many scientific and engineering problems do not naturally live on uniform Cartesian grids and why this matters for neural operators.

2.4.2 geo-FNO

Describe the geo-FNO approach and how it extends Fourier-based operator learning to non-uniform meshes.

Chapter 3

Experimental Setup

This chapter describes the datasets, evaluation metrics, and training procedures used throughout the thesis. It also clarifies how models with different output-resolution behavior are compared in the discretization transfer experiments.

3.1 Datasets

The experiments in this thesis use both uniform-grid and non-uniform datasets. For each dataset, the goal of this section is to describe the underlying physical problem, the mapping learned by the model, the data source, and the aspects of the dataset that are relevant for interpreting the results. Dataset-specific implementation details are included only when they affect the conclusions drawn later in the thesis. In particular, the full generation procedure for the self-generated Navier-Stokes dataset is deferred to Appendix X.

3.1.1 Uniform-grid datasets

The discretization invariance experiments on regular grids use two datasets: a self-generated Navier-Stokes dataset and a Darcy flow dataset obtained from prior work.

Navier-Stokes

One of the main datasets used in this thesis is a two-dimensional Navier-Stokes dataset generated specifically for this work. The dataset represents a PDE solution operator on a uniform grid and is therefore well suited for evaluating how neural operators behave when the discretization is changed between training and testing. In the main text, it is sufficient to state that the dataset was generated in-house, to describe the physical quantity provided as input and the field predicted as output, and to summarize the resolution(s) used in the experiments. The exact simulation pipeline, parameter ranges, and any numerical settings that may be relevant for reproducibility are better placed in an appendix and referenced here.

Because this dataset was generated specifically for the present study, it also plays an important role in interpreting later results. Any quirks of the data distribution that help

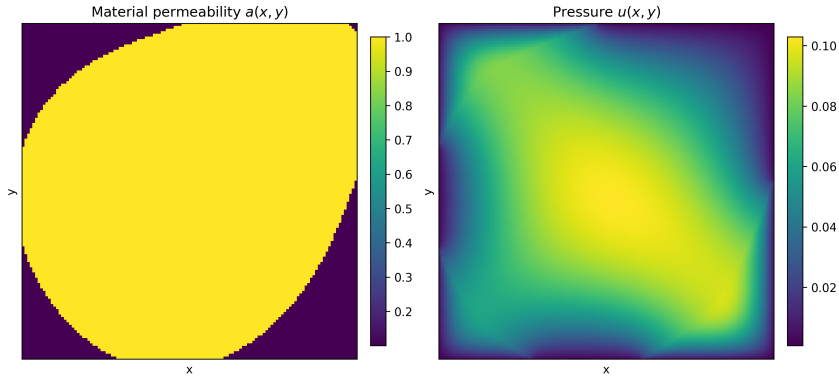

Figure 3.1: Example of an input diffusion coefficient field $a(x,y)$ and the resulting pressure field $u(x,y)$ with Dirichlet boundary conditions.

explain observed model behavior should therefore be mentioned briefly in this chapter, even if the full generation procedure is deferred.

Darcy flow

For the regular-grid experiments, we use the Darcy flow dataset introduced by Li et al.[1]. In this benchmark, the task is to learn the solution operator of a steady-state two-dimensional Darcy flow problem on the unit square:

$$-\nabla \cdot (a(x)\nabla u(x)) = f(x), \qquad x \in (0,1)^2, \tag{3.1}$$

$$u(x) = 0, \qquad x \in \partial(0,1)^2, \tag{3.2}$$

The input is a spatially varying diffusion coefficient field $a(x)$, and the output is the corresponding solution field $u(x)$ under a prescribed forcing term and Dirichlet boundary conditions. An example input-output mapping can be seen in figure 3.1. This PDE has many applications within physics and engineering, including drainage and irrigation in agricultural soils and fluid migration around nuclear waste repositories [2]. The dataset contains 10000 samples at a resolution of 128×128.

3.1.2 Non-uniform datasets

The experiments on irregular domains use three datasets: the elasticity benchmark used in the original geo-FNO work, an incompressible Euler dataset modeling flow past a cylinder, and a nuclear engineering dataset consisting of temperature profiles for a single HTGR column.

Elasticity

The elasticity dataset is included because it is one of the canonical benchmarks used to evaluate geo-FNO on non-uniform meshes. It provides a direct connection to the original

geo-FNO setting and therefore helps assess whether the implementation used in this thesis reproduces the expected behavior on an established test case.

In the dataset description, it is enough to explain the physical problem at a high level, describe the input and output quantities, and note that the data is defined on an irregular geometry rather than a uniform Cartesian grid. The main text should also cite the original source and report the experimental resolution or mesh representation at the level needed to interpret the results.

Incompressible Euler flow past a cylinder

A second non-uniform benchmark is an incompressible Euler dataset describing fluid flow past a cylinder. This dataset was used in prior work on operator learning for PDEs and is included here because it provides a fluid-dynamics test case on a geometry that is not naturally aligned with a regular grid.

For the purposes of this thesis, the most important information is what the model receives as input, what field it predicts, and why the domain geometry makes this dataset relevant for geo-FNO. As with the elasticity dataset, the source should be cited and the description should include the mesh or sampling representation used in the experiments. Fine-grained details of the original data-generation process are not needed unless they become important for later analysis.

HTGR column temperature dataset

The final non-uniform dataset consists of temperature profiles for a single column of a high temperature gas-cooled reactor (HTGR) and was generated by a colleague. This dataset is of particular interest because it connects the thesis to the motivating nuclear engineering application. Compared with the public benchmark datasets, it is also a more domain-specific test of whether geo-FNO can learn useful operators for reactor-related data.

In the main text, this dataset should be introduced in terms of its physical meaning, the variables that define the input and output, and the way spatial locations are represented. Since the dataset is not a standard public benchmark, it is also worth briefly explaining how it was obtained and why it is scientifically relevant. If there are strong structural regularities or simplifications in the data that later help explain the results, those should be noted here as well.

3.1.3 Scope note

In addition to the datasets discussed above, a non-uniformly sampled Navier-Stokes dataset was used during troubleshooting. Since it was not part of the final evaluation, it is not described as a primary dataset in this thesis.

3.2 Evaluation Metrics

Define the metrics used to compare models, such as the L_2 error and relative error, and explain how they are aggregated over the test set.

3.3 Training Details

Summarize the optimization settings, learning-rate schedule, early stopping or patience settings, number of epochs, and any preprocessing steps shared across experiments.

3.4 Resolution Transfer Protocol

This section should explain how discretization invariance is evaluated in practice.

3.4.1 Train-test resolution split

Explain that models are trained on one resolution and evaluated on other resolutions to measure generalization across discretizations.

3.4.2 Architectural comparison details

Clarify that FNO naturally predicts on the input resolution, whereas CNO predicts on its internal resolution, and explain how outputs are made comparable for a fair evaluation.

Chapter 4

Results

This chapter presents the empirical findings of the thesis. It is split into two parts: a study of discretization invariance for FNO and CNO, and an investigation of geo-FNO on non-uniform meshes, including its application to a nuclear dataset.

4.1 Part I: Discretization Invariance Study

4.1.1 FNO versus CNO

Experimental design

Describe the setup in which each model is trained on one grid resolution and evaluated on unseen resolutions.

Results

Present the quantitative comparison between FNO and CNO and discuss how performance changes under resolution transfer.

4.1.2 Interpretation of discretization effects

This section should connect the observations back to the theoretical differences between the models.

FNO

Discuss spectral truncation and the fact that Fourier mode index does not preserve the same physical meaning when the discretization changes.

CNO

Discuss the role of locality, the bandlimited assumption, and the implicit multiscale structure in producing steadier behavior across resolutions.

Connection to theory

Tie the empirical behavior back to the contrast between bandlimiting and Fourier truncation.

4.2 Part II: Non-uniform Meshes and geo-FNO

4.2.1 Results

Present the geo-FNO results on the irregular-domain datasets and discuss the overall performance.

4.2.2 Analysis of seam artifacts

This section can document the seam artifact, explain how it manifests in different datasets, and analyze why some datasets are more robust to it than others.

Observed artifact

Describe the seam, the associated corruption of low-k_x modes, and the evidence that it appears across datasets.

Tracing the source

Explain how the issue was traced to the `conv0` layer and what was ruled out during debugging, including the approximate `fft2d`/`ifft2d` functions and the handling of complex weights.

Patch and effect

Explain the bug in the original implementation, how it was fixed, and show the before-and-after effect of the patch.

4.2.3 Nuclear application

Describe the nuclear dataset, explain the prediction task, and discuss how geo-FNO performs on unseen z levels and temperatures.

Dataset characteristics

Explain why the dataset is relatively simple and consider supporting that claim with a PCA visualization.

Generalization results

Discuss the extent to which the experiment demonstrates that geo-FNO extends to nuclear applications.

4.3 Writing note

An open presentation choice is whether to show the original seam-affected plots in the main text before introducing the patch, or to present only the corrected results in the main text and move most debugging evidence to an appendix.

Chapter 5

Related Work

(Your topic here) is a well explored domain in computer science. This chapter discusses technologies related to (my research project). It is divided into ...

5.1 Related part I

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5.2 Related part II

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5.3 Related part III

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Chapter 6

Conclusions and Future Work

This chapter gives an overview of the project's contributions. After this overview, we will reflect on the results and draw some conclusions. Finally, some ideas for future work will be discussed.

6.1 Contributions

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6.2 Conclusions

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6.3 Discussion/Reflection

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6.4 Future work

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Appendix A

Glossary

In this appendix we give an overview of frequently used terms and abbreviations.

foo: ...

bar: ...

Appendix B

Requirements and Guidelines

This chapter details some requirements and guidelines for MSc theses submitted to the Software Engineering Research Group.

B.1 Requirements

B.1.1 Layout

- Your thesis should contain the formal title pages included in this document (the page with the TU Delft logo and the one that contains the abstract, student id and thesis committee). Usually there is also a cover page containing the thesis title and the author (this document has one) but this can be omitted if desired.
- Base font should be an 11 point serif font (such as Times, New Century Schoolbook or Computer Modern). Do not use sans-serif fonts such as Arial or Helvetica. *Sans-serif type is intrinsically less legible than seriffed type* [**?**].
- The final thesis and drafts submitted for reviewing should be printed double-sided on A4 paper.

B.1.2 Content

- The thesis should contain the following chapters:
  - Introduction.
    Describes project context, goals and your research question(s). In addition it contains an overview of how (the remainder of) your thesis is structured.
  - One or (usually) more "main" chapters.
    Present your work, the experiments conducted, tool(s) developed, case study performed, etc.
  - Overview of Related Work
    Discusses scientific literature related to your work and describes how those approaches differ from what you did.

- Discussion/Evaluation/Reflection
  What went well, what went less well, what can be improved?
- Conclusions, Contributions, and (Recommendations for) Future Work
- Bibliography

B.1.3 Bibliography

- Make sure you've included all required data such as journal, conference, publisher, editor and page-numbers. When you're using BIBTEX, this means that it won't complain when running `bibtex your-main-tex-file`.
- Make sure you use proper bibliographic references. This especially means that you should avoid references that **only** point at a website and not at a printed publication.

  For example, it's OK to add a URL with the entry for an article describing a tool to point at its homepage, but it's not OK to just use the URL and not mention the article.

B.2 Guidelines

- The main chapters of a typical thesis contain approximately 50 pages.
- A typical thesis contains approximately 50 bibliographic references.
- Make sure your thesis structure is balanced (check this in the table of contents).

  Typically the main chapters should be of equal length. If they aren't, you might want to revise your structure by merging or splitting some chapters/sections.

  In addition, the (sub)section hierarchies with the chapters should typically be balanced and of similar depth. If one or more are much deeper nested than others in the same chapter this generally signals structuring problems.
- Whenever you submit a draft of your thesis to your supervisor for reviewing, make sure that you have checked the spelling and grammar. Moreover, *read it yourself at least once from start to end, before submitting to your supervisor*.

  Your supervisor is not a spelling/grammar checker!
- Whenever you submit a second draft, include a short text which describes the changes w.r.t. the previous version.